G. NARAYANAMMA INSTITUTE OF TECHNOLOGY AND SCIENCE

(AUTONOMOUS) (For Women)

Shaikpet, Hyderabad - 500 104

II B.Tech. I Semester – II MID Examinations January-2024

Academic Year 2023-2024

SUBJECT: Design and Analysis of Algorithms
SUBJECT CODE: 123AN **Max Marks: 30**
BRANCH: CSE A, B, C, CSE(AI & ML), CSE(Data Science) **Time: 2Hrs**
Date: 18-1-2024 (AN)

Blooms Taxonomy Levels:
Level 1 – Remembering, Level 2 – Understanding, Level 3 – Applying, Level 4 – Analyzing, Level 5 – Evaluating, Level 6 - Creating

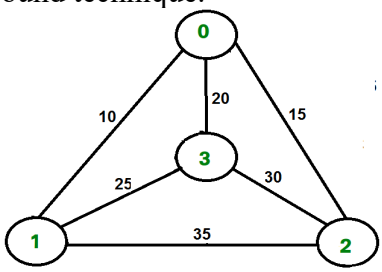
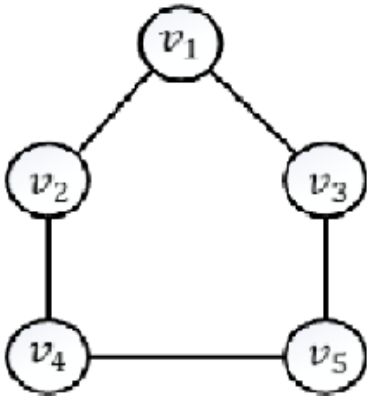
PART-A

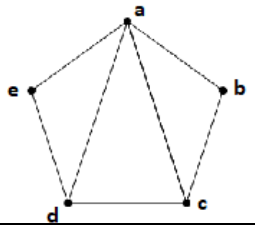
(Answer all the questions. Each question carries 1 mark) **1 x10 = 10 Marks**

S.No. 1	Question	CO	BT Level
a)	Which algorithmic design technique is used to solve reliability design problem?	CO4	L-1
b)	Define dominance rule.	CO4	L-1
c)	Give the time complexity of solving travelling sales person problem using dynamic approach.	CO4	L-2
d)	What are Explicit and Implicit constraints?	CO5	L-1
e)	Define an E-Node.	CO5	L-1
f)	List the three common search strategies used in branch and bound.	CO5	L-1
g)	What is an Approximation algorithm.	CO5	L-1
h)	Is P subset of NP? Justify the statement.	CO6	L-2
i)	List the functions used for writing non- deterministic algorithms.	CO3	L-1
j)	Give the equation to represent the cost function ‘c ’ satisfying the triangle inequality of TSP problem for all vertices u,v, w Є V	CO6	L-1

PART-B

(Answer 04 full questions. Each question carries 5 marks) **4 x 5 = 20 Marks**

S.No.	Question	Marks	CO	BT Level
2.a)	Obtain an OBST with equal probabilities for the identifier set (a1,a2,a3)=(if, stop, while)	[3M]	CO4	L-3
b)	State weather the Knapsack problem is maximization problem or minimization problem? Justify by elaborating on the problem statement.	[2M]	CO4	L-2
3.a)	Compare and contrast Backtracking with Branch and Bound technique.	[2M]	CO5	L-2
b)	Draw the portion of the state space tree generated by the following knapsack instance. n=5, (p1, p2, p3, p4. p5) = (10,15,6,8,4), (w1, w2,w3,w4,w5)=(4,6,3,4,2), m=12.	[3M]	CO5	L-3
4.a)	Give the Control abstraction of LC Search strategy.	[2M]	CO5	L-1
b)	Solve the travelling salesman problem for the following graph using branch and bound technique. 	[3M]	CO5	L-3
5.a)	Explain the approximation algorithm to find the vertex cover of a graph, find the minimum vertex cover for the following graph. 	[3M]	CO6	L-2
b)	Define P, NP, NP-Hard, NP-Complete classes and show the relationship between them with a neat sketch.	[2M]	CO6	L-1

6.a)	What is triangle-inequality and explain how it is used to solve travelling sales man problem.	[3M]	CO6	L-1
b)	Give the deterministic and Non- deterministic algorithms for sorting a set of integers.	[2M]	CO3	L-1
7.a)	How many minimum numbers of colors are needed to color the following example graph? Show their assignment by tracing the backtracking algorithm for Graph coloring. 	[3M]	CO4	L-3
b)	Prove that satisfiability is in P iff P=NP.	[2M]	CO6	L-1